Richa S Dhawale

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CURRENT AFFILIATION

Postdoctoral Researcher at Centre for Hydrology (Global Institute of Water Security-GIWS), at the University of Saskatchewan, Canada

(October 2022- Present)

- Developing a comprehensive vulnerability index integrating socio-economic, climatic, and hydrological data.
- Developing a framework and a workshop for stakeholders on Social Vulnerability Index
- Validating agent-based models of household and community flood response through behavioral datasets.

EDUCATION

Indian Institute of Technology Kharagpur, Kharagpur **2017- 2022**
Ph.D at Architecture and Regional Planning (ARP)
Thesis: A system dynamics framework to assess the urban drought.

- Developed a dynamic modeling approach to simulate urban water stress and policy trade-offs, with applications in climate adaptation and risk management.
- Conducted extensive primary surveys, interviewing government officials, planners, and community members to capture multi-level perspectives on drought risk and adaptive practices.
- Published peer-reviewed articles on drought indices, risk assessment, and remote sensing applications.

Indian Institute of Technology Kharagpur, Kharagpur **2015- 2017**
Masters in City Planning (MCP) at Architecture and Regional Planning (ARP); CGPA 8.42 /10.0
Thesis: Planning for Drought: Moving from crisis Management to Risk Assessment, Maharashtra.

- Specialized in climate-resilient planning and water governance, with a focus on integrating risk assessment into regional development policies.
- Carried out primary surveys and stakeholder interviews with local officials and communities to assess perceptions, preparedness, and planning gaps in drought management.

Visveswaraya National Institute of Technology, Nagpur **2010-2015**
Bachelors in Architecture (B. Arch) CGPA 8.02/10.0
Thesis: Tribal Museum and Learning Centre, Nagpur.

- Trained in urban and regional planning, spatial design, and sustainable infrastructure, with a focus on socio-cultural and environmental dimensions of planning.
- Conducted urban planning surveys to understand settlement patterns, community needs, and socio-cultural aspects of design within regional development contexts.

PROFESSIONAL ENAGAGEMENT

Reviewer – Fellowship Programme, Coalition for Disaster Resilient Infrastructure (CDRI) (2024 & 2025 Cohorts)

- Conducted double-blind peer reviews of fellowship proposals on wildfires, coastal resilience, energy, extreme heat, and disaster risk reduction.
- Supported global research efforts in resilience and adaptation through international collaboration.

Technical Expert / Reviewer – Public Safety Canada, Community Risk Index (CRI) Research Task Team

- Serving as subject-matter expert in disaster resilience, hazard and risk modeling, and emergency management.
- Reviewing methodology and providing technical guidance for the development of the Community Risk Index (CRI), a national-level disaster risk data product.

Panel Member – Water Security Panel, Engineering Design Mutualisms Laboratory (August 2025)

- Invited to present perspectives on coupled human–water systems and share expertise on behavioral dimensions of water security in the Saskatchewan River Basin and Delta.
- Engaged with participants, enriching discussion on human–water interactions and resilience strategies.

Invited speaker- Water Resource Center (WRC), Department of Civil Engineering, National Institute of Engineering, Mysuru 2021.

- Delivered a lecture on Applications of Remote Sensing in Disaster Management.
- Engaged with 50+ undergraduate and graduate students, introducing satellite-based monitoring and its role in risk assessment.
- Shared practical insights on using geospatial tools for flood and drought analysis

Invited Speaker in **Sri Sri University, Bhubaneshwar on Disaster Management** in August 2019

- Conducted a session with 4th and 5th year architecture students on Disaster Management of Droughts.
- Illustrated the integration of urban planning, policy, and geospatial methods in building resilience.
- Encouraged discussion on the role of architects and planners in addressing water-related hazards.

Smart City Project, Varanasi – Collaboration with Georgia Tech (Atlanta) & IIT Varanasi (2019)

- Participated in a 10-day field study in Varanasi focused on environmental planning and urban development.
- Conducted ground surveys, stakeholder interactions, and data collection for infrastructure assessment.
- Co-authored reports and delivered presentations with the project team on findings and recommendations.

TEACHING EXPERIENCE

Institute of Technology Kharagpur (*Teaching Assistant)* TA for the following Subjects:

- **Industrial Development Cooperation, Navi Mumbai (CIDCO) (*Internship*)** *May-Jul 2016*

Climatology and Solar Architecture	July 2020 - Nov 2020, July 2018 - Nov 2018, July2019 – Nov 2019
Architectural Design- II	Jan 2019 - Apr 2019
Computer-Aided Design and Simulation (AutoCAD and Revit)	Jan 2018 – Apr 2018
Architectural Design Studio- I	Jul 2016 – Nov 2016

National Programme on Technology Enhanced Learning (NPTEL) course on Geo Spatial Analysis in Urban Planning 2020 conducted by IIT Kharagpur and organized by IIT Madras

- Prepared and analyzed Kharghar and Panvel Nodal Plan. Preparation of
- Redevelopment plans for Navi Mumbai.

ADDITIONAL WORK EXPERIENCE

- **Architects' Combine, Mumbai (*Internship*)** *May-Dec 2014*
  - Worked on project design of Tata Institute of Fundamental research, Hyderabad. Interior detailing of a seminar hall, Green Technology Building, University of Mumbai.
  - Preparation of drawing schedule for GKR, Chennai.
  - Preparation of rendered drawing for a competition project of IIT Gandhinagar. Worked on the projects like Pattiswaram (Tamil Nadu), Award offset factory (Silvasa), SIES College (Nerul) Preparation of 3D models for Karunya University, Coimbatore.
- **Karani and Sons Consultants, Mumbai (*Internship*)** *May-Jul 2013*
  - Architecture planning and public layout as per BMC DC and MCGM rules and regulations. Preparation of area statements.
  - Space planning and designing.

SKILLS

Computer: Programming with R, GIS frameworks, SPSS, Anylogic, Vensim, Stella, AutoCAD, Python, Hydrologic Model
Language: English, Hindi, Marathi

PUBLICATIONS

- **Dhawale, R.**, Schuster, CJ., & Pietroniro, A (2024). Assessing the multidimensional nature of flood and drought vulnerability index: A systematic review of literature. *In International Journal of Disaster Risk Reduction* (Vol. 112). Elsevier Ltd. https://doi.org/10.1016/j.ijdrr.2024.104764.
- **Dhawale, R.**, Paul, S. K. & George JS. (2022). Water Balance Analysis Using Palmer Drought Severity Index for Drought-Prone Region of Marathwada, India. *In the Journal of River Basin Management* https://doi.org/10.1080/15715124.2022.2079661.
- **Dhawale, R.,** Paul, S. (2019). Moving from Crisis Management to Risk Assessment for Drought Planning Using Standardized Precipitation Index (SPI) and Standardized Groundwater Level Index (SWI): Case Study of Marathwada, India. *In the Global Assessment Report (GAR) published by United Nations International Strategy for Disaster Reduction (UNISDR) Global Platform for Disaster Risk Reduction*.
- **Dhawale, R.**, & Paul, S. K. (2018). A comparative analysis of drought indices on vegetation through remote sensing for Latur region of India. *ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *XLII-5*(November), 403–407. https://doi.org/10.5194/isprs- archives-XLII-5-403-2018.
- Addai, O., Soares, C., **Dhawale, R**., Schuster, CJ., Spiteri, R. (2025) Flood-ABM: An agent-based model of differential flood effects on population groups and their decision-making processes, International Journal of Disaster Risk Reduction, Volume 128, 105698, ISSN 2212-4209, https://doi.org/10.1016/j.ijdrr.2025.105698.
- George JS, Paul SK**, & Dhawale R**. (2021) Multilayer network structure and city size: A cross-sectional analysis of global cities to detect the correlation between street and terrain. Environment and Planning B: *Urban Analytics and City Science.* August 2021. doi: 10.1177/23998083211039853.
- George J, Paul SK, & **Dhawale R.** (2021) A cellular-automata model for assessing the sensitivity of the street network to natural terrain, *Annals of GIS*, 27:3, 261-272, DOI: 10.1080/19475683.2021.1936173
- George J, Paul SK, & **Dhawale R.** (2021) Contrasting urban greenness across cities with varying trends in above-normal weather events, *Nature-Based Solutions*, DOI: https://doi.org/10.1016/j.nbsj.2021.100008.

CONFERENCE PAPERS

- **Dhawale, R.**, Aghaie, V., Schuster, CJ., & Pietroniro, A. (2023). Assessing Flood Vulnerability in the Saskatchewan River Basin: A Multi-Dimensional Approach. *Global Water Futures Finale (May 2023).*
- **Dhawale, R.**, & Paul, S. K. (2018). A comparative analysis of drought indices on vegetation through remote sensing for Latur region of India. *ISPRS - International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, *XLII-5*(November), 403–407. https://doi.org/10.5194/isprs- archives-XLII-5-403-2018
- **Dhawale, R**., Paul, S., & George, J. (2019). Analyzing the water balance for drought prone regions of Marathwada, India. In J. Andharia, R. Sinha, & P. D. Chakrabarti (Eds.), *World Disaster Management Congress (Compendium of abstract)* (p. 264). Mumbai, ISBN: 978-93-5346-919-1.
- **Dhawale, R.**, Paul, S., Roy, S., George, J., & Kumar, A. (2019). Assessing the occurrence of drought based on SPI, NDVI and LST for Indian region. In *European Geosciences Union*. Vienna. (April 2019)
- Roy, S., Paul, S. K., **Dhawale, R**., Kumar, A., & Agnihotri, V. (2019). Empowering Children: Using GIS framework to locate and allocate schools in Indian cities to increase active school transportation. In *European Geosciences Union*. Vienna (April 2019).
- George, J., Paul, S. K., **Dhawale, R**., Patnaik A. (2019). Natural Terrain and Urban form: Interpreting locational patterns of arterial roads and activity centres. In *7th Annual International Conference on Architecture and Civil Engineering (ACE)*. Singapore. Print ISSN: 2301-394X, E-Periodical ISSN: 2301- 3958

- Chetia, L., Paul, S. K., **Dhawale, R**., & Joy, N. M. (2022). Identification and Mapping of 2019 Flood Extents Using Sentinel-1. A Images: A Case of Barpeta District, Assam. *In Sustainable Water Resources Management* (pp. 165-173). Springer, Singapore

POSTER PRESENTATION:

- Soares, C., Addai, O., **Dhawale, R**., Spiteri, R., Clark, M., Lloyd-Smith, P., & Schuster, CJ. (2024). Operationalizing Equity in Coupled Human-Water Systems Modeling. World Water Day, Global Institute of Water Security, University of Saskatchewan (March 2024).
- Soares, C., Kolen, K., **Dhawale, R.**, Aghaie, V., Zoll, J., Spiteri, R., & Schuster, CJ. (2024). Intrapersonal and Interpersonal Determinants of Household and Community Disaster Mitigation Behaviour: A scoping Review Protocol. World Water Day, Global Institute of Water Security, University of Saskatchewan (March 2024).

COURSES/ WORKSHOPS:

- Course on IWRM for climate resilience by United Nation, 2021 (6 weeks).
- Symposium on Urban Informatics and AI-driven Analytics, Department of Architecture and Regional Planning, IIT Kharagpur 2018
- 16th IIRS-ISRO Outreach program on Geospatial Technology in Urban Planning, February 2016 (Duration 2 months).
- City development plan of Kalyani & Gayeshpur, West Bengal. (Transport and Traffic Planning) 2016 Urban space system design – Paradip, Orissa, 2015

NOTED FELLOWSHIPS AND AWARDS

MHRD Fellowship for Teaching Assistantship during the MCP Program. (2015-2017)

MHRD Fellowship for Teaching Assistantship during the PhD Program. (2017-2022)